

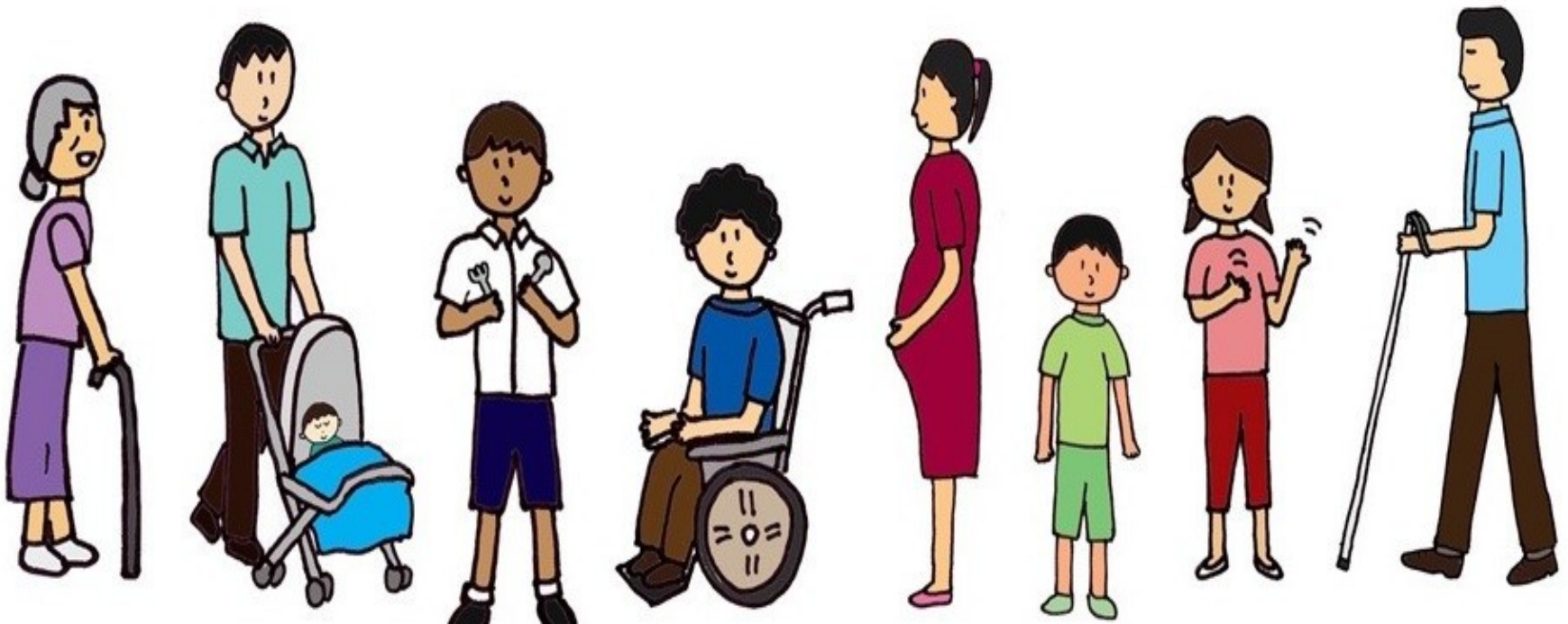
Human Computer Interaction

Universal Design and User Support

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Universal design

- *the process of designing products* that are accessible by all users in all circumstances, taking account of human diversity in disabilities, age and culture

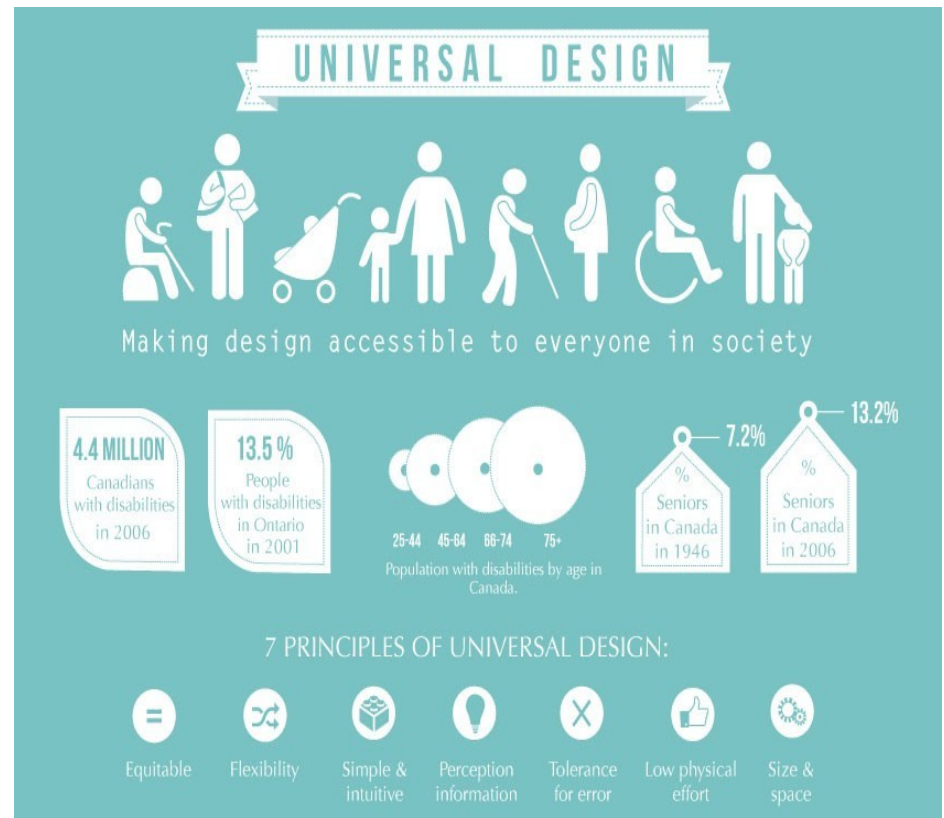


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Seven Universal Design Principles

- In 1990s, *North Carolina State University in the USA* proposed seven universal design principles

1. *Equitable use*
2. *Flexibility in use*
3. *simple and intuitive to use*
4. *Perceptible information*
5. *Tolerance for error*
6. *Low physical effort*
7. *Size and space for approach and use*



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Seven Universal Design Principles Con...

1. ***Equitable use:*** the design is useful to people with a range of abilities and appealing (tempting) to all.

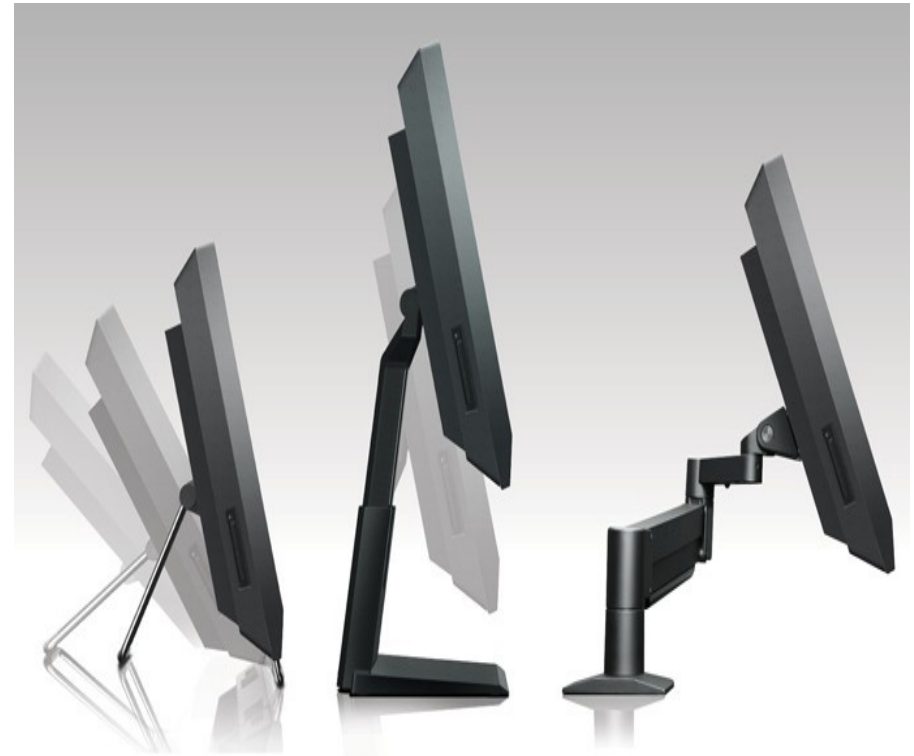


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Seven Universal Design Principles

Con...

2. ***Flexibility in use:*** the design is adaptively to the user's pace (speed), precision (accuracy), and custom(habit)



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Seven Universal Design Principles

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3. The system be **simple and intuitive (perceptive) to use**, regardless of the knowledge, experience, language or level of concentration of the user



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Seven Universal Design Principles

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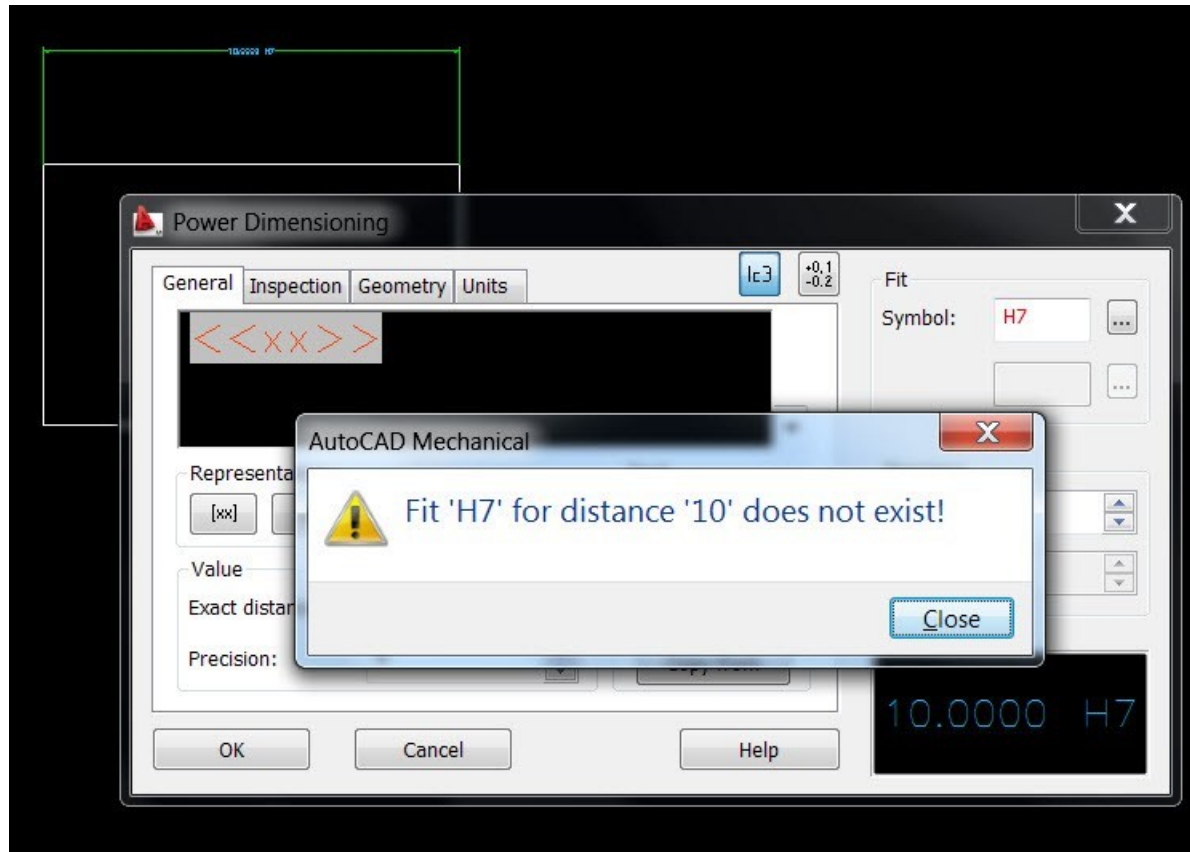
4. **Perceptible (observable) information:** The design should provide effective communication of information regardless of the environmental conditions or the user's



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Seven Universal Design Principles Con...

5. ***Tolerance for error:*** minimizing the impact and damage caused by mistakes or unintended behavior



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6. ***Low physical effort:*** systems should be designed to be comfortable to use, minimizing physical effort and fatigue (tiredness)



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Seven Universal Design Principles Con...

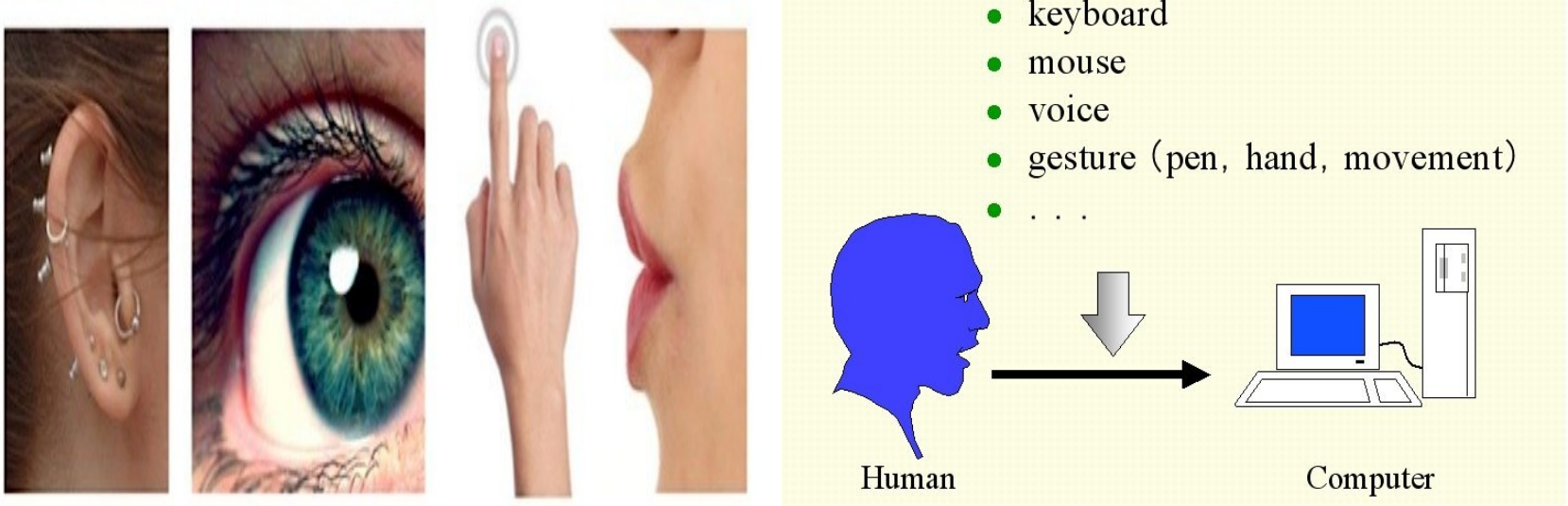
7. ***Size and space for approach and use:*** The placement of the system should be reached and used by any user regardless of body size, posture or mobility (ie. seated or standing users)



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Multi-modal Interaction

- provide access to system information and functionality through a range of *different input and output channels*
- The 5 senses (sight, sound, touch, taste and smell) are used by us every day and provide a fuller interaction with the natural world
- Computers rarely offer such a rich interaction



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Designing for Diversity

The designer considers three key areas:

1. Designing for users with disabilities

- visual impairment
- hearing impairment
- physical impairment
- speech impairment
- dyslexia
- autism

1. Designing for different age groups

- Old people
- Children

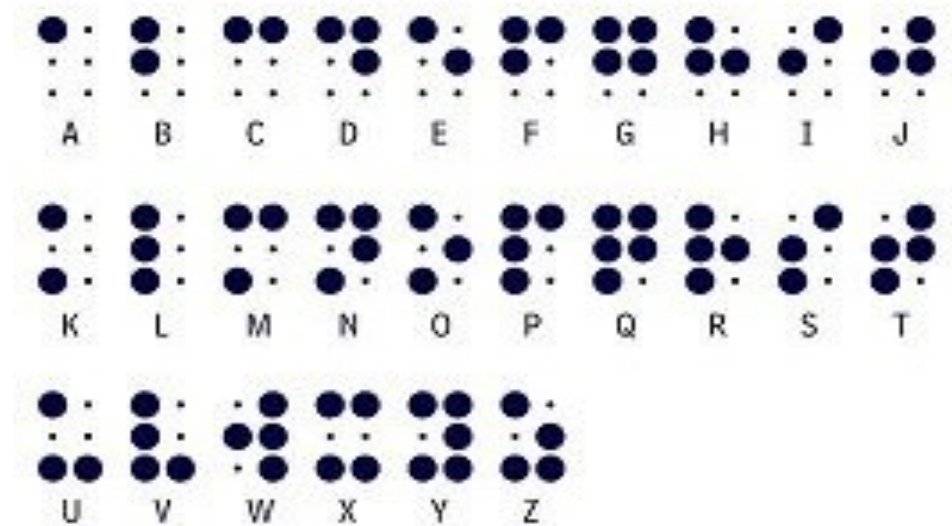
1. Designing for cultural differences

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Designing for Diversity Con...

visual impairment

- The rise in the use of graphical interfaces reduces the possibilities for visually impaired users



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Designing for Diversity Con..

hearing impairment

- have little impact on the use of an interface
- Captioning audio content making audio files easier and more efficient to index and search

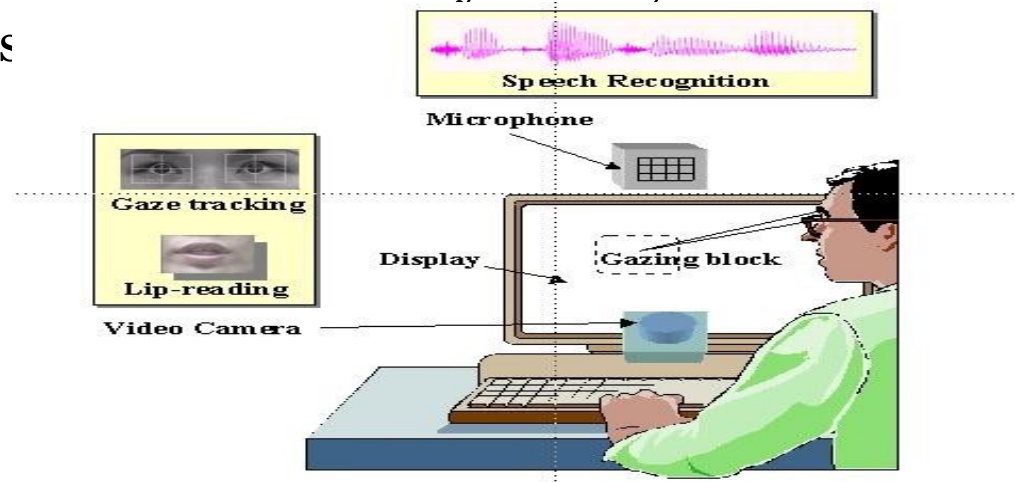


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physical impairment

- Users with physical disabilities vary in the amount of control and movement over their hands, impact in mouse control difficulty
- **eyegaze system** which tracks eye movements to control the cursor,
- keyboard driver that can be attached to the user's head.
- predictive system, such as the **Reactive keyboard**, cut the typing requirement cons



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- **speech impairment**

multimedia systems provide a number of tools for communication, including synthetic speech and text-based communication and conferencing systems



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Dyslexia

- Users with cognitive disabilities such as dyslexia can find textual information difficult



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Designing for Diversity Con...

Autism

- affects a person's ability to communicate and interact with people around them and to make sense of their environment

1. Social interaction – problems
2. Communication – problems
3. Imagination – problems

How might universal design of technology assist people with autism?

communication and education.



Designing for Diversity Con...

Designing for different age groups

- **Older people:**

email and instant messaging, can provide social interaction in cases where lack of mobility or speech difficulties and reduce face-to-face possibilities.

- **Children**

Information in Graphics, sound and text, Pen-based interfaces, touch and voice interfaces for children than keyboard and mouse



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Designing for Diversity Con...

Designing for cultural differences

- The other factors such as age, gender, race, sexuality, class, religion and political persuasion, may all influence an individual's response to a system
- The designer practice universal design which include language, cultural symbols, gestures and use of color



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User Support

- Even if an interactive system is properly designed, the user will require varies assistance at various times, dependent on many factors:
 - their familiarity with the system,
 - the job they are trying to do, and so on.

There are four main types of assistance that

1. quick reference
2. task-specific help
3. full explanation
4. tutorial



User support is provided by different types of support system

- i. Help systems are problem oriented and specific
- ii. Documentation is system oriented and generic

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User Support Con...

- ***Quick reference***
 - used as a reminder to the user for the details of tools
 - Example: find a particular command option and the syntax of the command
- ***Task-specific help***
 - required how to apply the tool to his particular problem
- ***Full explanation***
 - used when the more experienced or inquisitive (questioning) user may require a full explanation of a tool or command
- ***Tutorial***
 - particularly aimed at new users of a tool and provides step-by-step instruction (perhaps by working through examples) of how to use the tool

Requirements of user support

What features will the help system have? These features are:

- ***Availability***

- access help at any time during his interaction with the system.
- Ideally, run concurrently with main application.

- ***Accuracy and completeness***

- provided should be accurate and complete.
- If provided proves not to match the actual behavior of the system the user will become disillusioned (disappointed) with the help facilities.

- ***Consistency***

- Online help should also be consistent with paper documentation in terms of content, terminology and style of presentation

Requirements of user support Con...

- ***Robustness***

- When the system is behaving unexpectedly or has failed altogether ,the help system should be correct error handling and predictable behavior.

- ***Flexibility***

- allow each user to interact with it in a way appropriate to his needs, experience and task.

- ***Unobtrusiveness***

- The help system should not prevent the user from continuing with normal work.
- The textual help system on a non-windowed interface may interrupt the user's work.
- A possible solution to this is split-screen presentation.

Approaches to user support

- A number of different approaches to providing help, each of which meets a particular need.
- vary from simple captions to full adaptive help and tutoring systems.

The styles of help are given below:

• ***Command assistance***

- The user requests help on a particular command and is presented with a help screen or manual page describing it.
- e.g., UNIX man, DOS help, Windows Help.
- It is good for quick reference

• ***Command prompts***

- In command line interfaces, command prompts provide help , usually in the form of correct usage prompts when the user encounters an error.

• ***Context sensitive help***

- used in menu-based systems to provide help on menu options
- Example: The Microsoft Office tool-tips , Web page rollover.

Approaches to user support Con...

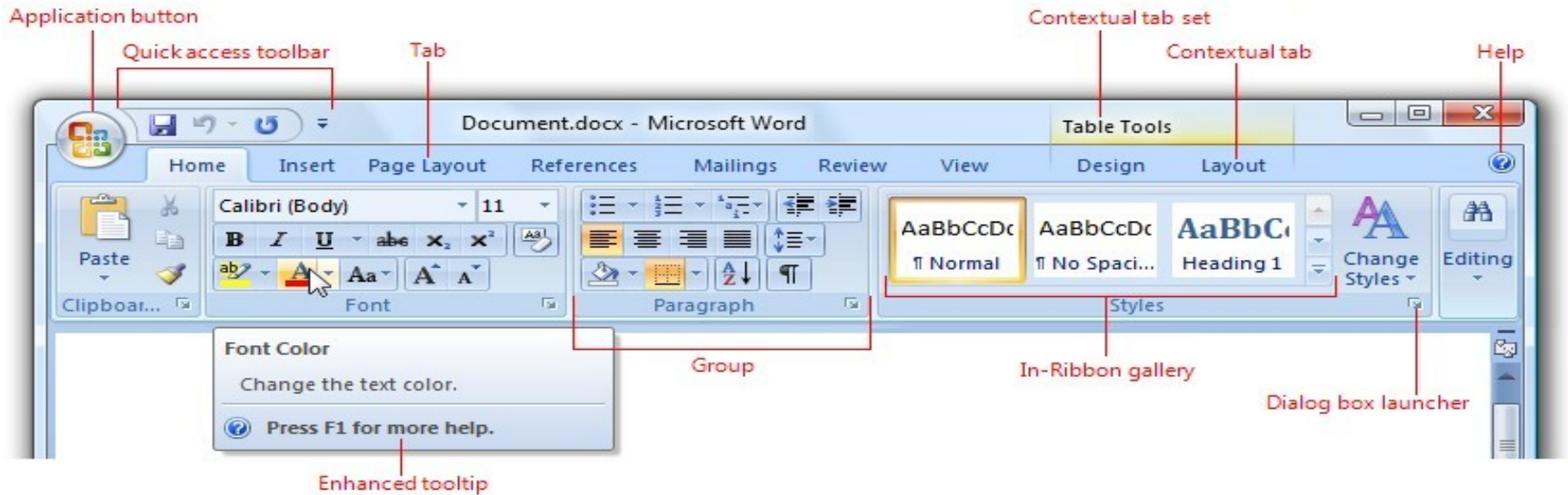


Table of codes (authorities)

- Alphabetical listing
- Shortcut (alt+shift+I)
- Hide codes (ctrl+shift+8)

• Insert table of authorities

		to be able to have a little bit more space so
level 1 codes		
Childhood		childhood/adulthood
Countryside		and childhood/adulthood
Nice and like		it
Returning happily		



Add categories
e.g. Level 1, Level 2,
interpretative
repertoires, canonical
narratives

Approaches to user support Con...

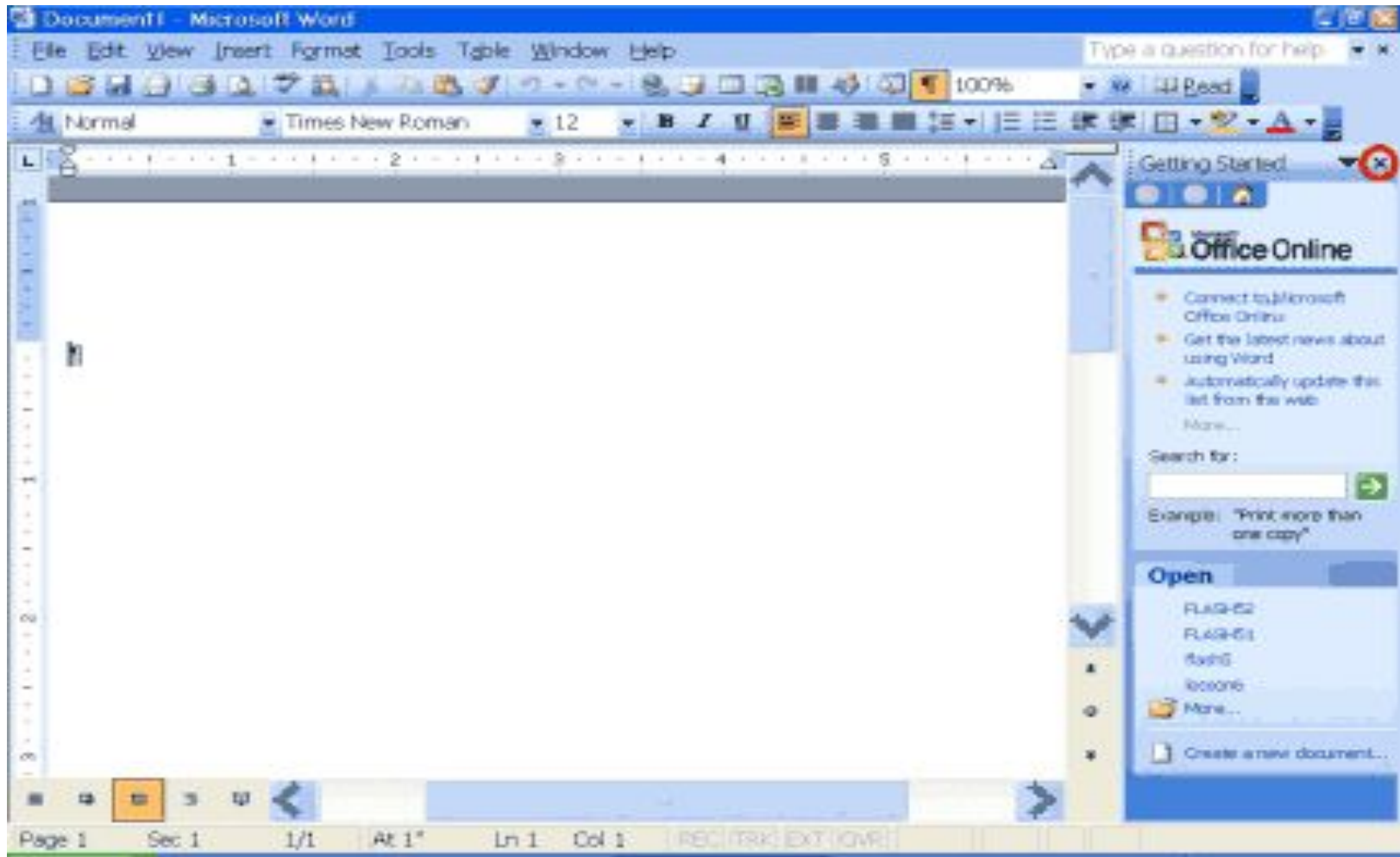
- *On-line tutorials*

- The user get how the application works by **experimenting with examples and demonstration of how to perform a task**
- The user can **repeat parts of the tutorial** if needed.

- *On-line documentation*

- **paper documentation available** on computer.
- **material available continually** (assuming the machine is running!) in the same medium to a large number of users concurrently.
- difficult to browse.
- Documentation structured using **hypertext supports browsing**

Approaches to user support Con...



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Approaches to user support Con...

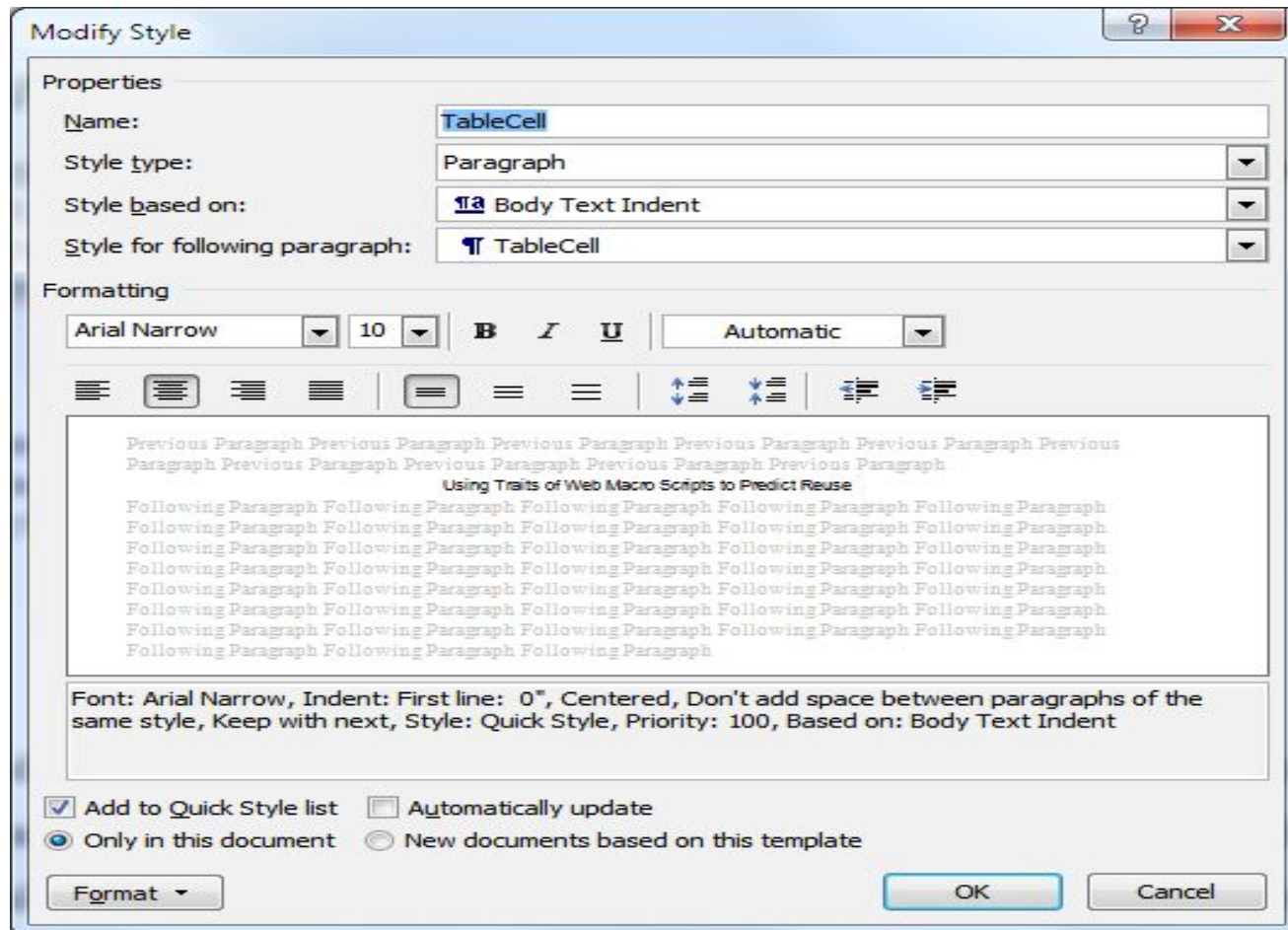
- ***Wizards***

- A *wizard* is a **task-specific tool**.
- allow the user actually to complete the tasks safely, quickly and accurately.
- Example, the **Microsoft Word resumé**.
- allow the user to move back a step as well as forward, will provide a progress indicator showing how much of the task is completed and how many steps remain, and will offer sufficient information to allow the user to answer the questions

- ***Assistants***

- software tools that **monitor user behavior** and offer suggestions or hints when they recognize familiar sequences.
- Example : **Eager**, a software agent. When it notices the user repeating a sequence of actions, a **cat icon appears**, suggesting the next action

Approaches to user support Con...



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Adaptive help systems

- In any large or complex computer system, users will be familiar with a subset of the available
 - functionality, demonstrating expertise in
 - some applications having no experience with
 - others,
unaware of their existence.
- attempt to address these above problems by *adapting the help to the individual user* and by actively *suggesting alternative courses* to the user may not be aware.
- *Intelligent systems* include domain-specific expert systems, intelligent tutoring systems and general adaptive interfaces.

Adaptive help systems

Other Resources

- **Excel Tutorials on the Internet**

- <http://www.fgcu.edu/support/office2000/excel/>
- <http://www.baycongroup.com/el0.htm>
- <http://www.studyfinance.com/lessons/excel/>
- <http://www.usd.edu/trio/tut/excel/>

- **Useful Powerpoint help**

- <http://www.actden.com/pp/print.htm>
- http://www.ellenfinkelstein.com/powerpoint_tip.html
- http://www.masterviews.com/2001/11/15/chart_types_how_to_select_appropriate_chart_types.htm

- **Microsoft Word Tutorials**

- <http://www.tutorialbox.com/tutors/off2000/word/>
- <http://www.baycongroup.com/wlesson0.htm>
- <http://www.public.iastate.edu/~hschmidt/wordtutorial.html>

Designing User Support Systems

There are a number of things which the designer should take into account.

1. User support is not an 'add on'
 - should be designed integrally with the system.
2. Concentrate on content and context of help rather than technological issues.
3. Make decisions about how the help will be presented to the user and how this will be affected by implementation issues.

Designing User Support SystemsCon...

Presentation issues

- How is help requested?
 - command, button, function (on/off), separate application
- How is help displayed?
 - new window, whole screen, split screen,
 - pop-up boxes, hint icons
- Effective presentation requires
 - clear, familiar, consistent language
 - instructional rather than descriptive language
 - avoidance of blocks of text
 - clear indication of summary and example information

Designing User Support SystemsCon...

Implementation issues

1. Will help be an *operating system command, a meta-command or an application*?
2. What *physical constraint* does the machine impose *in terms of screen space, memory capacity and speed*?
3. How the help data is to be *structured: in a single file, a file hierarchy, a database*? any structure should be flexible and extensible
4. Will users be able to *browse through the system* or only request help on one topic at a time?
5. Will the user make a *hard copy of part of the help system* to study later (manuals and documentation).
6. The designer should *consider the authors of help material as well as its users*. Even if the designer *writes the initial help texts*, these *will be extended* by other authors at different times.

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